Variations in verb-stranding VP-ellipsis: verbal identity and phases
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Abstract: The main goal of this paper is to establish a novel generalization regarding two well-known but poorly understood types of variation in ellipsis constructions. The first type concerns identity conditions imposed on the stranded verb in verb-stranding VP-ellipsis constructions and the second type concerns extraction possibilities out of ellipsis targets. While these two types of variation appear on the surface to be unrelated, using a typologically diverse range of data, I show that they are correlated. I show that the underlying factor which regulates both types of variation is the phasal status of the ellipsis target in such constructions. In VSVE constructions where the ellipsis target is a phase verbal identity is required and extraction is not possible; when the ellipsis target is a phase complement verbal identity is not enforced and extraction is allowed. Further, I develop an analysis that explains this correlation in these phasal terms.
Keywords: verb-stranding VP-ellipsis, extraction, phases, cross-linguistic variation

1 Introduction

The empirical focus of this paper is two well-known but poorly understood types of cross-linguistic variation regarding ellipsis. The first type concerns verb-stranding VP-ellipsis (VSVE) constructions. It been observed that languages vary regarding whether they allow for VSVE, which involves elision of the predicate phrase (i.e. vP/VP) but with the survival of the main verb. For example, while English does not allow VSVE (1); Bosnian/Croatian/Serbian (BCS) does (2).[1]

(1) a. * Mark doesn't write papers carefully but Maria writes Δ.
b. Mark doesn't write papers carefully but Maria does Δ.

(2) Marko ne piše rad pažljivo, a Marija piše ~~rad pažljivo~~.
Marko not writes paper carefully, but Marija writes paper carefully
'Marko isn't writing the paper carefully, but Maria is.'
(Stjepanovic 1999: 43)

[1]Unless cited, all the data in this paper are from the author's consultation with native speakers of the relevant language.

Among VSVE languages, there is further variation. In some languages, strict lexical identity is imposed between the antecedent and stranded verb. For example, VSVE in BCS, as in (2,3). In other languages, the antecedent and stranded verbs can be distinct. For example, VSVE in European Portuguese, as in (4). Here, the stranded verb *ofereceu* is distinct from antecedent main verb *vendeu*.

(3) *Ivan piše rad pažljivo, a njegov asistent čita Δ.
Ivan writes paper carefully, and his assistant reads
'Ivan is writing a paper carefully and his assistant is reading (the paper carefully).'

(4) O João vendeu livros á Teresa ontem e a Ana ofereceu
the João sold books to.the Teresa yesterday and the Ana offered
~~livros á Teresa ontem~~.
books to.the Teresa yesterday
'João sold books to Teresa yesterday and Ana offered them.'
(Santos 2009: 57)

The second type of variation concerns extraction out of ellipsis targets. Across ellipsis constructions, there is well-known variability regarding the possibility of extraction out of the ellipsis target (Aelbrecht 2010; van Craenenbroeck and Lipták 2010; Bošković 2014, among others). To illustrate, while English permits *wh*-extraction out of a TP-ellipsis target (as in (5)), i.e. sluicing, Japanese clausal ellipsis does not allow extraction out of the elided clause, as illustrated in (6).

(5) Martha claimed Peter yelled at someone but I don't know who$_i$ [$_{TP}$ Δ].

(6) *Sono hon-o$_i$ Taroo-wa [$_{CP}$ Hanako-ga t$_i$ katta to] itta ga, zassi-o$_j$
that book-ACC Taroo-TOP Hanako-NOM bought C said though magazine-ACC
Ziroo-wa [$_{CP}$ Δ] itta.
Ziroo-TOP said
'That book, Taro said that Hanako bought and, that magazine, Ziroo said (Hanako bought)'
(Saito 2007: 210)

Although both types of variation— verbal identity requirements and extraction possibilities— are well-known, it is generally thought that these two types are unrelated to each other. That is, they are uncorrelated types of variation regarding ellipsis.

The main goal of this paper is to show that these two types of variation are, in fact, correlated and that there is a tight relationship between whether a given VSVE construction enforces verbal identity and whether it allows for extraction out of the VSVE target. Based on a number of VSVE languages, many of which are typologically unrelated, I will show that in VSVE contexts where verbal identity is enforced, Ā-extraction out of the VSVE target is not possible; in contexts where verbal identity is not enforced, Ā-extraction is allowed. Further, I develop an analysis of this generalization that connects these two types of surface variation in terms of a single underlying difference regarding the phasal status of the ellipsis target. I argue that in VSVE constructions where the ellipsis target is a phase complement, verbal identity can be violated and Ā-extraction is possible. But in VSVE constructions where the ellipsis target is a full phase, verbal identity is imposed and Ā-extraction is blocked.

The paper is structured as follows. In section 2, I outline the VSVE phenomenon and the variation observed regarding verbal identity, which I argue can be traced to a difference in head-dependencies involved in VSVE. In section 3, I discuss the second type of variation regarding extraction. Here, I show that whether extraction out of VSVE is possible correlates with the identity conditions on the stranded verb. In section 4, I show how this correlation is due to the phasal status of the VSVE target and develop an analysis of this correlation in these terms. Section 5 concludes.

2 The verb stranding VP-ellipsis phenomenon

I will first outline the basic contours of the VSVE phenomenon and what distinguishes it from other, similar processes. Following this discussion, I turn to the first type of variation: whether identity is imposed on the stranded verb in VSVE contexts.

VSVE is a phenomenon where the predicate phrase, i.e. vP/VP, and predicate-internal elements (i.e. internal arguments and adjuncts) are elided, but the main verb is pronounced. This phenomenon is illustrated below with an instance of VSVE in Uyghur. In (7), the predicate phrase is targeted for ellipsis, where predicate-internal elements, i.e. object and adjunct, are elided but the main verb *berdi* is pronounced.

(7) Men Ayghülga sowghatni ettigende berdim, anam=mu Δ
1SG Ayghül.DAT gift.ACC morning.LOC give.PST.1SG mother.1POSS=also
berdi.
give.PST.3SG
I gave Ayghül a gift in the morning, my mother gave (Ayghül a gift in the morning) too.'
(Jenkins, 2025)

As discussed in a number of previous studies, (e.g. in Park 1997; Oku 1998b; Goldberg 2005; Gribanova 2013b; Landau 2018, 2020, 2023; Simpson 2023), VSVE can produce surface strings that are superficially similar to the output of other processes, namely, null argument *pro* (8) or (multiple) argument ellipsis (9). Subsequently, it is crucial to distinguish VSVE from these other processes.

(8) [… [$_{vP}$ [$_{VP}$ *pro* V] v] …]

(9) [… [$_{vP}$ [$_{VP}$ ~~DP~~ V] v] …]

First, VSVE cases can be distinguished from cases of null argument *pro* due to the availability of both strict and sloppy interpretations of the elided argument under VSVE, which is not possible with a *pro* argument (Otani and Whitman, 1991; Oku, 1998b; Goldberg, 2005; Saito, 2007; Takahashi, 2008; Şener and Takahashi, 2010; Sakamoto, 2020). As illustrated in (10a) with Uyghur VSVE, the null DP can receive both a strict and sloppy interpretation (i.e. *Ayghül's bag* or *Zemire's bag*) indicating that the DP is elided, rather than being a *pro* argument (which should only support a strict interpretation; compare with (10b)).

(10) a. Ayghül somkisini qachilaydu, lëkin Zemire Δ qachilamaydu.
Ayghül bag.3POSS.ACC pack.PRS.3SG but Zemire pack.NEG.NPST.3SG
'Ayghül packs her bag, but Zemire won't pack (Ayghül's/Zemire's bag).'

b. Ayghül somkisini qachilaydu, lëkin Zemire uni qachilamaydu.
Ayghül bag.3POSS.ACC pack.PRS.3SG but Zemire 3SG.ACC pack.NEG.NPST.3SG
'Ayghül packs her bag, but Zemire won't pack it (= Ayghül's/*Zemire's bag).'
(adapted from Jenkins 2025)

Second, potential VSVE cases must be distinguished from cases of (multiple) argument ellipsis, i.e. (9). A standard diagnostic to distinguish between cases of VSVE and argument ellipsis is whether the ellipsis site can admit of an 'adjunct-inclusive' interpretation (Park, 1997; Oku, 1998b; Goldberg, 2005). The logic of this diagnostic is that given that argument ellipsis can only target arguments and not non-argument adjuncts, if the ellipsis clause allows for an adjunct-inclusive interpretation, then the ellipsis target must be large enough to include both the arguments and the adjoined phrase, i.e. the vP/VP. To illustrate this diagnostic, in Uyghur adverbs (11a) and secondary predicates (11b) cannot be independently elided.

(11) a. Zemire Ayghülga da'im alma berdi. Nilufar=mu Ayghülga alma berdi.
Zemire Ayghül.DAT often apple give.PST.3SG Nilufar=also Ayghül.DAT apple give.PST.3SG
'Zemire often gave apples to Ayghül. Nilufar (*often) gave apples to Ayghül too.' (Jenkins 2025)

b. Zemire ma yataquyni qizil boyaldi, Ayghül a yataquyni boyaldi.
Zemire this room.ACC red paint.PST.3SG Ayghül that room.ACC paint.PST.3SG
'Zemire painted this room red and Ayghül painted that room (*red).'

However, in VSVE contexts, as in (12a) and (12b), the elided adjuncts can be recovered.

(12) a. Zemire Ayghülga da'im alma berdi. Nilufar=mu Δ berdi.
Zemire Ayghül.DAT often apple give.PST.3SG Nilufar=also give.PST.3SG
'Zemire often gave apples to Ayghül. Nilufar (often) gave (apples to Ayghül) too.' (Jenkins, 2025)

b. Zemire yataquyini qizil boyaldi lëkin Ayghül Δ boyalmadi.
Zemire room.3POSS.ACC red paint.PST.3SG but Ayghül paint.NEG.PST.3SG
'Zemire painted her room red, but Ayghül didn't paint (her room red).'

That the adjuncts can be recovered in (12a,12b) but not in (11a,11b), indicates that the relevant ellipsis process in (12a,12b) targets a phrase large enough to include both internal arguments and adjuncts, i.e. the vP/VP.

Further, it is important to highlight that the observation that the ellipsis clauses in (12a, 12b) admit of an adjunct-inclusive interpretation cannot be explained in terms either 'adjunct ellipsis' (Oku, 2016; Landau, 2020) or other, pragmatic factors (Landau, 2018, 2020, 2023). Fujiwara (2025) shows that such factors can be controlled for by testing whether (or not) the elided vP/VP, which includes the adjunct, can be directly denied in a discourse or targeted by a propositional anaphor. It is well-known that while semantically represented content, which in turn must be syntactically present, can be directly denied or targeted by a propositional anaphor in a discourse, pragmatic content cannot be (for discussion see Tonhauser 2012; Snider 2017; Koev 2018). To

illustrate, in the dialogue in (13), Bill replies to Peter's question with the answer in (13b). Crucially, the felicity of (13b) depends on the pragmatic inference that *Martha wasn't the designated driver* given that *Martha drank a bottle of wine*, which is not semantically encoded in Bill's response in (13b). Subsequently, since this information is not semantically represented, it cannot be directly denied or targeted with the anaphor *that*, thus, the infelicity of Ernie's response in (13c) (compare with the dialogue in (14), where the proposition *Martha was the designated driver* is semantically represented represented in (14b)).

(13) *Context: Peter, Bill, and Ernie are trying to figure out who drove everyone home from the party last night.*
   a. Peter: Who was the designated driver last night? Wasn't it Martha?
   b. Bill: Well, I saw Martha drink an entire bottle of wine.
   c. Ernie: # I don't believe *that* (= Martha wasn't the designated driver)! I swore I saw her volunteering to drive.

(14) a. Peter: Who was the designated driver last night? Wasn't it Martha?
   b. Bill: Yes, Martha was the designated driver.
   c. Ernie: I don't believe *that* (= Martha was the designated driver)! I saw her drinking wine all night.

Fujiwara (2025) argues that if VSVE involves elision of a vP/VP, including the adjunct, then direct denial of the adjunct-inclusive interpretation should be possible and also targetable by an anaphor due to the adjunct being syntactically, and, thus, semantically present. As shown below, this is the case for Uyghur VSVE. In (15a), where the direct object is present but not the adjunct, the only available interpretation is that *Ayghül painted the room (some color or another)*. Crucially, the adjunct-inclusive interpretation is not available, i.e. *Ayghül painted the room red*. Moreover, the response in (15b) is infelicitous. In contrast, in the VSVE construction in (16a), where an adjunct-inclusive interpretation of the ellipsis clause is available, the response in (16b) is felicitous and can be construed as directly disputing whether Ayghül painted the room red or not. Further, the vP/VP (including the secondary predicate) is targetable by the anaphor *bu* indicating that the secondary predicate must be syntactically present in the ellipsis site in (16a).

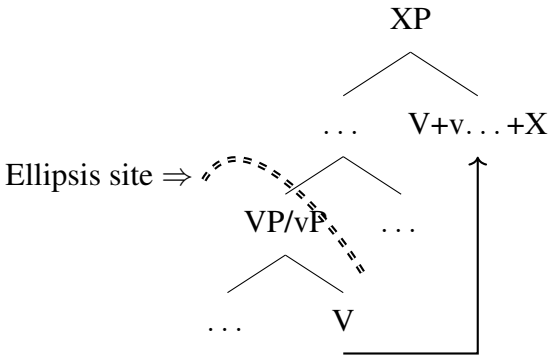
(15) a. Zemire ma yataquyni qizil boyaldi, Ayghül a yataquyni boyaldi.
   Zemire this room.ACC red paint.PST.3SG Ayghül that room.ACC paint.PST.3SG
   'Zemire painted this room red and Ayghül painted that room.'
   b. #Bu yalran. Ayghül yataquyni yeshil boyaldi.
   that false Ayghül room.ACC green paint.PST.3SG
   'That's (=Ayghül painted that room red) not true. Ayghül painted the room green.'

(16) a. Zemire yataquyini qizil boyaldi lëkin Ayghül Δ boyalmadi.
   Zemire room.3POSS.ACC red paint.PST.3SG but Ayghül paint.NEG.PST.3SG
   'Zemire painted her room red, but Ayghül didn't paint (her room red).'
   b. Bu yalran. Ayghül=mu yataquyni qizil boyaldi.
   that false Ayghül=also room.ACC red paint.PST.3SG
   'That's (=Ayghül didn't paint her room red) not true. Ayghül painted her room red too.'

Given that an adjunct-inclusive interpretation of the ellipsis clause in (16a) can be directly denied and targeted by a propositional anaphor, this confirms that the relevant process in Uyghur VSVE involves vP/VP ellipsis and cannot be due to pragmatic factors (cf. Landau 2018, 2020, 2023). Further, that there is a felicity contrast between the responses in (15b) and (16b) indicates that Uyghur VSVE cannot be due to argument plus adjunct ellipsis (cf. Oku 2016; Landau 2020). If adjuncts could be independently elided, then an adjunct construal of the second clause in (15a) should be available and, crucially, it should be felicitous to directly deny such a construal (contrary to fact, cf. (15b)). Thus, given that elements such as adverb and secondary predicates cannot be independently elided but can be elided in VSVE contexts indicates that the VSVE phenomenon is distinct from both cases of object *pro* and (multiple) argument ellipsis constructions.

Given these distinctive features of VSVE, previous works (e.g. in Otani and Whitman 1991; Doron 1999; Goldberg 2005; Lipták 2012; Gribanova 2013b; Merchant 2018; see also Huang 1987, 1988, 1991) have proposed that VSVE involves two components. Namely, elision of the predicate phrase (i.e. vP/VP) with displacement of V outside of the ellipsis target. This process is sketched below in (17). As (17) illustrates, V moves to a functional projection, XP, outside of the VP-ellipsis target, which elides elements within the VP, i.e. adjuncts and internal arguments.

(17)



Cross-linguistically, VSVE has been attested in a range of languages, many of which are typologically unrelated. As was shown, Uyghur is one such language (as in (7)). Further languages where VSVE has been attested include: Hindi-Urdu (Simpson et al., 2013; Manetta, 2019; Simpson, 2023), European Portuguese (Cyrino and Matos, 2002, 2005; Santos, 2009; Martins, 2016), Greek (Merchant, 2018), Japanese (Funakoshi, 2012, 2014, 2016; Fujiwara, 2017, 2022, 2025; Fenger, 2020), Bosnian/Serbian/Croatian (BCS) (Stjepanovic, 1999; Todorovic, 2016), Swahili (Ngonyani, 1996, 1998), and Russian (Gribanova, 2013a,b, 2017).[2]

As shown below in (18a), in Hindi-Urdu manner adverb adjuncts cannot be independently elided (Manetta, 2019). But in VSVE contexts, as in (19a), an adjunct-inclusive interpretation of the ellipsis clause is available, thus, indicating that the predicate phrase is the ellipsis target (where internal arguments and adjuncts are elided). Moreover, as the continuation in (19b) indicates, this material can be directly denied and referred to by a propositional anaphor, thus, indicating that the adverb is syntactically present in the ellipsis site (cf. (18b)).

(18) a. Ram-ne Chomsky-ka naya lekh dhyaan-se paRh-a magar Raj-ne
Ram-ERG Chomsky-GEN new writing carefully read-PFV.M.SG but Raj-ERG

[2]There are additional VSVE languages which will be put aside in this study due to being unable to independently verify all the additional data and tests that will be discussed in this paper for these languages.

nahiiN naya lekh paRh-a.
NEG new writing read-PFV.M.SG

Ram read the new paper by Chomsky carefully, but Raj did not read the new paper (*carefully). (Manetta 2019: 925)

b. #aisa nahiiN hai. Raj bahut dhyaan-se Chomsky-ke arguments paRh rahaa thaa.
this NEG AUX.PRS Raj very carefully Chomsky-GEN arguments read PROG.M AUX.PST.M.

'This is not true. Raj was reading Chomsky's arguments very carefully.'

(19) a. Ram-ne Chomsky-ka naya lekh dhyaan-se paRh-a magar Raj-ne Δ nahiiN paRh-a.
Ram-ERG Chomsky-GEN new writing carefully read-PFV.M.SG but Raj-ERG NEG read-PFV.M.SG

Ram read the new paper by Chomsky carefully, but Raj did not read (the new paper by Chomsky carefully). (Manetta 2019: 924)

b. aisa nahiiN hai. Raj bahut dhyaan-se Chomsky-ke arguments paRh rahaa thaa.
this NEG AUX.PRS Raj very carefully Chomsky-GEN arguments read PROG.M AUX.PST.M.

'This is not true. Raj was reading Chomsky's arguments very carefully.'

The same contrast is observed in European Portuguese. As shown below, (20a) cannot be felicitously responded to with (20b), thus, indicating that the manner adverb adjunct cannot be independently elided. But under VSVE, as in (21a), not only is an adjunct interpretation available but (21b) is a felicitous response, thus further confirming that (21a) is an instance of VSVE.

(20) a. A Raquel não limpou o carro cuidadosamente. Mas a Ana limpou o carro.
the Raquel NEG cleaned the car carefully but the Ana cleaned the car

'Raquel didn't clean the car carefully but Ana cleaned the car.'

b. #Eu não acredito nisso, a Ana é muito relaxada quando ela limpa, havia migalhas de pão espalhaadas no assento do carro.
I NEG believe this the Ana is very relaxed when she cleans was crumbs of bread scattered on.the seat of.the car

'I don't believe this, Ana is very sloppy when she cleans, there were crumbs all over the car seat.'

(21) a. A Raquel não limpou o carro cuidadosamente. Mas a Ana limpou Δ.
the Raquel NEG cleaned the car carefully but the Ana cleaned

'Raquel didn't clean the car carefully but Ana cleaned (the car carefully).'

b. Eu não acredito nisso, a Ana é muito relaxada quando ela limpa, havia migalhas de pão espalhaadas no assento do carro.
I NEG believe this the Ana is very relaxed when she cleans was crumbs of bread scattered on.the seat of.the car

'I don't believe this, Ana is very sloppy when she cleans, there were crumbs all over the car seat.'

Greek also displays the relevant contrast indicating that VSVE is an operative process (see Merchant 2018 for additional evidence that Greek instantiates VSVE). As shown in (22a) the adjunct cannot be independently elided, as the infelicity of the continuation in (22b) indicates. However, in VSVE contexts, an adjunct-inclusive interpretation of the gap is possible (23a) and can be felicitously responded to with (23b).

(22) a. O Petros den diavase prosektika kanena vivlio tou Chomsky, alla o Kostas
the Petros NEG read completely any book of Chomsky but the Kostas
diavase ena vivlio tou Chomsky.
read a book of Chomsky.
'Petros didn't completely read a book by Chomsky, but Kostas read a book by Chomsky.'

b. #Den to pistevo afto. O Kostas den diavase prosektika kanena vivlio tou Chomsky.
NEG it believe that the Kostas NEG read completely any book of Chomsky
'I don't believe that. Kostas didn't completely read a book by Chomsky.'

(23) a. O Petros den diavase prosektika kanena vivlio tou Chomsky, alla o Kostas
the Petros NEG read completely any book of Chomsky but the Kostas
diavase Δ.
read
'Petros didn't completely read a book by Chomsky, but Kostas read (completely a book by Chomsky).'

b. Den to pistevo afto. O Kostas den diavase prosektika kanena vivlio tou Chomsky.
NEG it believe that the Kostas NEG read completely any book of Chomsky
'I don't believe that. Kostas didn't completely read a book by Chomsky.'

Japanese can also be shown to instantiate VSVE. Fujiwara (2025) shows that adjuncts cannot be independently elided due to the inability of propositional anaphors to target potential missing adjuncts, as (24a,24b) shows. However, in VSVE contexts, as in (25a), (25b) is a felicitous response, thus, indicating the presence of an elided adjunct and that the entire vP/VP is the ellipsis target.[3]

[3] As suggested by an anonymous reviewer, it is worth noting that there is debate regarding the existence of Japanese VSVE (in addition to related processes, i.e. null object *pro* (Hoji, 1998) and argument ellipsis (Oku 1998a,b; Goldberg 2005; Saito 2007; Takahashi 2008, 2014; Şener and Takahashi 2010; Maeda 2019; Sakamoto 2020, *i.a.*). In particular, there are works that argue that Japanese lacks VSVE and that cases like (25a) are instances of argument ellipsis, where the adjunct-inclusive interpretation is due to either the adjunct being independently elided (Oku, 2016; Kobayashi, 2020; Landau, 2020; Tanabe and Kobayashi, 2023) or a pragmatic inference (Landau, 2023). However, Fujiwara (2025, p.c.) argues that neither the adjunct ellipsis nor pragmatic inference proposals can account for the contrast in (24a) and (25a), where (24b) is not a felicitous response but (25b) is felicitous. The issue is that if an adjunct-inclusive interpretation were due to adjunct ellipsis, then an adjunct-inclusive interpretation in (24a) should be available and the missing adjunct should be targetable by the propositional anaphor in (24b) (contrary to fact). Conversely, if an adjunct-inclusive interpretation were due to a pragmatic inference, then the the missing adjunct in (25a) should not be targetable by the propositional anaphor in (25b). Due to thus contrast, Fujiwara (2025) concludes there is strong evidence that (25a) is due to VSVE.

(24) a. John-wa teineini kotti-no kuruma-o arat-ta yo. Bill-wa atti-no kuruma-o
John-TOP carefully this-GEN car-ACC wash-PST SFP Bill-TOP that-GEN car-ACC
arat-ta yo.
wash-PST SFP
'John washed this car carefully. Bill washed that car.'

b. #Sore-wa uso da ne. Datte Bill-wa teinei zyanakatta kara.
that-TOP lie COP SFP because Bill-TOP careful was.not because
'That's a lie. Because Bill wasn't careful.' (Fujiwara, 2025)

(25) a. John-wa teineini kuruma-o arat-ta yo. Bill-mo Δ arat-ta yo.
John-TOP carefully car-ACC wash-PST SFP Bill-also wash-PST SFP
'John washed a car carefully. Bill also washed (a car carefully).'

b. Sore-wa uso da ne. Datte Bill-wa teinei zyanakatta kara.
that-TOP lie COP SFP because Bill-TOP careful was.not because
'That's a lie. Because Bill wasn't careful.' (Fujiwara, 2025)

VSVE has also been attested in BCS (Stjepanovic, 1999; Todorovic, 2016). As shown below, that BCS instantiates VSVE is confirmed by the observation that while an adjunct-inclusive interpretation is unavailable in (26a); such an interpretation is available in (27a). Moreover, that (27a) can be felicitously replied to with (27b) further confirms that the relevant process is VSVE (cf. (26b)).

(26) a. Maro ne piše rad pažljivo, a Marija piše rad.
Marko not writes paper carefully but Marija writes paper
'Marko isn't writing a paper carefully and Maria is writing a paper.'

b. #To ti ne vjerujem. Marija piše rad nemarno.
that you not believe Marija writes paper carelessly
'I don't believe you. Maria is writing a paper carelessly.'

(27) a. Marko ne piše rad pažljivo, a Marija piše ~~rad pažljivo~~.
Marko not writes paper carefully, but Marija writes paper carefully
'Marko isn't writing the paper carefully, but Maria is.'
(Stjepanovic 1999: 43)

b. To ti ne vjerujem. Marija piše rad nemarno.
that you not believe Marija writes paper carelessly
'I don't believe you. Maria is writing a paper carelessly.'

Swahili also allows for VSVE. As shown, while adjuncts cannot be independently elided, as (28a) and (28b) show, in VSVE contexts, such elements can be elided, as in (29a). Additionally, that the VSVE construction in (29a) admits of an adjunct-inclusive interpretation and can be replied to with (29b) indicates that the underlying process involved in (29a) is VSVE.

(28) a. Mw-alimu a-li-nunu-a ki-tabu ch-a Chomsky ili a-jifunz-e
1-teacher 1.SM-PST-buy-FV 7-book 7-a Chomsky in.order 1SM-learn-SUBJ
isimu, na wa-nafunzi wa-li-nunu-a ki-tabu ch-a Chomsky pia.
linguistics, and 2-student 2.SM-PST-buy-FV 7-book 7-a Chomsky too
'The teacher bought Chomsky's book in order to learn linguistics and the students bought Chomsky's book too.'

b. #Hapana, hiyo si kweli. Wa-nafunzi wa-li-nunu-a ki-tabu ch-a Chomsky
no that is.not true 2-student 2.SM-PST-buy-FV 7-book 7-a Chomsky
ili wa-som-e mambo ya siasa.
in.order 2.SM-study-SUBJ matters of government
'No, that isn't true. The students bought Chomsky's book to study politics.'

(29) a. Mw-alimu a-li-nunu-a ki-tabu ch-a Chomsky ili a-jifunz-e
1-teacher 1.SM-PST-buy-FV 7-book 7-a Chomsky in.order 1SM-learn-SUBJ
isimu, na wa-nafunzi wa-li-nunu-a Δ pia.
linguistics, and 2-student 2.SM-PST-buy-FV too
'The teacher bought Chomsky's book to study linguistics, and the students bought (Chomsky's book to study linguistics) too.

b. Hapana, hiyo si kweli. Wa-nafunzi wa-li-nunu-a ki-tabu ch-a Chomsky
no that is.not true 2-student 2.SM-PST-buy-FV 7-book 7-a Chomsky
ili wa-som-e mambo ya siasa.
in.order 2.SM-study-SUBJ matters of government
'No, that isn't true. The students bought Chomsky's book to study politics.'

As shown below, Russian also exhibits VSVE. While adjuncts cannot be independently elided (30a), an adjunct inclusive interpretation is available in (31a), which can be directly disputed, as in (31b), thus, indicating VSVE.

(30) a. Ivan ne prigotovil tort tochno sleduya retseptu, a Piter prigotovil
Ivan NEG prepared.M.SG cake exactly following recipe but Peter prepared.M.SG
tort.
cake
'Ivan did't prepare a cake exactly following the recipe, but Peter prepared a cake.'

b. #Eto nepravda. Piter redko sleduyet retseptu.
that not.true Peter rarely follows recipe
'That's not true. Peter rarely follows the recipe.'

(31) a. Ivan ne prigotovil tort tochno sleduya retseptu, a Piter prigotovil
Ivan NEG prepared.M.SG cake precisely following recipe but Peter prepared.M.SG
Δ.

'Ivan didn't prepare a cake precisely following the recipe but Peter prepared (a cake precisely following the recipe).'

b. Eto nepravda. Piter redko sleduyet retseptu.
that not.true Peter rarely follows recipe
'That's not true. Peter rarely follows the recipe.'

As the above cases show, in VSVE contexts the ellipsis clauses admit of an interpretation that is inclusive of both internal argument and adjuncts, thus, indicating that the ellipsis target is vP/VP despite the main verb being pronounced.[4]

[4]As suggested by an anonymous reviewer, it should be noted that among some of the above languages there is

While all of the above languages have been observed to exhibit VSVE, among them there is a typological split. In a number of works, it has been observed that for many VSVE languages, the stranded verb must be lexically identical to the verb in the antecedent clause (McCloskey, 1991; Goldberg, 2005). This is the case for VSVE in Hindi-Urdu (32) (Simpson et al., 2013; Manetta, 2019), Japanese (33) (Funakoshi 2014, 2016; Fenger 2020), Russian (34) (Gribanova, 2013b), and BCS (35). As shown in the examples below, when the stranded verb is distinct from the lexical verb, an adjunct-inclusive interpretation of the ellipsis clause is not possible. Given the unavailability of an adjunct-inclusive interpretation, this indicates that the underlying process cannot involve vP/VP-ellipsis, which involves elision of the both the internal argument(s) and adjuncts, which in turn should allow for an adjunct-inclusive interpretation of the ellipsis clause (contrary to fact).[5]

(32) *Ram-ne match shuru hone-se pehle pursii **uthaa**-ya lekin Raj-ne Δ
Ram-ERG match start become.INF-ABL before chair.F lift-PFV.F but Raj-ERG
giraa-yi.
drop-PFV.F
'Ram lifted a chair before the match started, but Raj dropped (the chair before the match started)'

(33) *Taro-wa (itumo) kurezittokaado-de hon-o **ka**-u kedo, Hanako-wa Δ
Taro-TOP (always) credit.card-with book-ACC buy-PRES but Hanako-TOP

some debate regarding whether that language admits of VSVE quite generally (in addition to null object *pro* and/or argument ellipsis) given that some speakers reject adjunct-inclusive interpretations in putative VSVE constructions. For relevant discussion see Goldberg 2005; Bailyn 2017; Landau 2020, 2021, 2023; also see fn. 3.

[5]It should be noted that for certain VSVE languages an additional diagnostic may be available which indicates whether (or not) verbal identity is imposed in VSVE. Namely, whether verbal mismatches are possible inside of an island. This diagnostic exploits the fact that in some languages, e.g. Hindi-Urdu, null object *pro* is not available inside of islands, while VSVE is (Manetta, 2019). Thus, the only process which can derive the surface string in (ib.) is VSVE (and not *pro*-drop). However, as shown in (iib.) when the verb inside the island is distinct from the antecedent, an overt object pronoun is required and VSVE is not possible, thus, indicating the Hindi-Urdu enforces verbal identity for VSVE.

i. a. Meena-ne nayii gaaRii aaj kharid-ii thii.
Meena-ERG new.F car.F today buy-PFV.F AUX.F
'Meena bought a new car today.'

b. kyaa aap yeh baat jaan-te haiN ki Manu-ne bhi Δ kharidii thii?
Q 2PL that fact know-HAB.PL AUX that Manu-ERG also buy-PFV.F AUX.F
'Did you know the fact that Manu also bought (a new car today)?' (Manetta 2019: 927)

ii. a. Sita-ne ghar-ko beech-a. Aap jaan-te haiN yeh baat ki Ram-ne ?*(us-ko) bhi
Sita-ERG house-ACC sell-PFV.M 2PL know-HAB.PL AUX.PL the fact that Ram-ERG 3SG-ACC also
banaay-aa thaa?
build-PFV.M AUX.M
'Sita sold the house. Did you know that Ram also built it?' (Manetta 2019: 934)

While this diagnostic is applicable in some VSVE languages, e.g. Hindi-Urdu, it is not applicable for all VSVE languages due to many languages being able to license object *pro* inside of islands (e.g. Landau (2020) shows that in some contexts European Portuguese may allow for *pro* inside of islands; cf. Raposo 1986). Given that lack of universal applicability of this island test for verbal identity, I set it aside for the present study.

koonyuusi-na-i.
purchase-NEG-PRES

'Taro (always) buys a book with a credit card, but Hanako (always) does not purchase a book with a credit card.'
(Funakoshi 2014: 329)

(34) a. Nepravitel'stvennye organizacii dolžny byli **razdavat'**
nongovernmental organizations supposed.PL were.PL out-give.2IMPV.INF
butylki vody bežencam na Gaiti.
bottles.ACC water.GEN refugees.DAT on Haiti.

'NGOs were supposed to be giving out bottled water to refugees in Haiti.'

b. #Nas očen' volnujut sluxi čto oni **prodavali** Δ.
us.ACC very worry.3PL rumors that they.NOM through-give.2.IMPV.PL

'The rumors that they were selling (bottled water to refugees in Haiti) really worry us.'
(Gribanova 2013b: 124)

(35) *Ivan **piše** rad pažljivo, a njegov asistent **čita** Δ.
Ivan writes paper carefully, and his assistant reads

'Ivan is writing a paper carefully and his assistant is reading (it carefully).'

In contrast to the above languages, in some VSVE languages the stranded verb can be lexically distinct from the main verb of the antecedent clause. This is the case in Uyghur VSVE. As shown in (36,37), the main verbs in the antecedent clauses are *berdi* and *aldi* but the stranded verbs are *ewetti* and *oghrildi*, respectively. Moreover, the gap in the ellipsis clauses admit of adjunct-inclusive interpretations (Jenkins, 2025).

(36) Zemire Ayghülga sowghatni ettigende **berdi**, lëkin Nilufar Δ **ewetti**.
Zemire Ayghül.DAT gift.ACC morning.LOC give.PST.3SG but Nilufar send.PST.3SG

'Zemire gave Ayghül a gift in the morning, but Nilufar sent (Ayghül a gift in the morning).'
(Jenkins, 2025)

(37) Zemire dukandin bir alma **aldi**, lëkin Ayghül Δ **oghrildi**.
Zemire store.ABL one apple buy.PST.3SG but Ayghül steal.PST.3SG

Zemire bought an apple from the store, but Ayghül stole (an apple from the store).'

Santos (2009) shows that European Portuguese also permits verbal mismatches in VSVE contexts. As shown in (38,39), the verbs in the antecedents and ellipsis clauses are distinct. Moreover, as noted by Santos (2009) the gap in the ellipsis clause in (38) admits of an adjunct-inclusive interpretation, thus, indicating elision of the vP/VP.

(38) O João **vendeu** livros á Teresa ontem e a Ana **ofereceu**
the João sold books to.the Teresa yesterday and the Ana offered
~~livros á Teresa ontem~~.
books to.the Teresa yesterday

'João sold books to Teresa yesterday and Ana offered them.'
(Santos 2009: 57)

(39) Os dossiers têm de ser **organizados** pela Ana mas, os dossiers dos impostos, ela
the files must be organized by.the Ana but the files of.the taxes she
não **quer** Δ.
NEG wants
Lit. 'The files must be organized by Ana, but those files concerning taxes, she doesn't want (to organize).' (Santos 2009: 103)

Greek also allows for verbal mismatches in VSVE contexts (Merchant, 2018). As (40,41) illustrate, the ellipsis clause admits of an interpretation that includes the PP-adjuncts indicative of VSVE.

(40) O Petros dhen **vrike** kanena vivlio sti vivliothiki, ute o Kostas **exase** Δ.
the Petros not found no book in.the library neither the Kostas lost
'Petros didn't find any book in the library, neither did Kostas lose (any book in the library).

(41) O Kostas **agorase** ena santouits apo tin kafeteria, alla o Petros **eklepse** Δ.
the Kostas bought a sandwich from the cafeteria, but the Petros stole
'Kostas bought a sandwich from the cafeteria, but Petros stole (a sandwich from the cafeteria).'

Finally, Swahili allows for verbal mismatches in VSVE contexts, as in (42,43). As illustrated in (42), the stranded verb is distinct from the antecedent verb and an adjunct-inclusive interpretation of the gap is available (i.e. the clausal-adjunct), indicating VSVE is the operative process.

(42) Peter a-na-**penda** kahawa kabla ya kw-enda kazi-ni, lakini Maria
Peter 1.SM-PRES-like 9.coffee before of to-go work-LOC but Maria
a-na-**chukia** Δ.
1.SM-PRES-hate
'Peter loves coffee before going to work, but Mary hates (coffee before going to work).'

(43) a. m-kurugenzi a-li-**omb**-a wazee wa-tembele-e ki-wanda ch-ote
1-director 1SM-PST-ask-FV them 2.SM-visit-SUBJ 7-factory 7-all
'The director told them to visit the entire factory.'

b. lakini meneja a-li-**amuru** ~~wa-tembele-e ki-wanda ch-ote~~.
but 1.manager 1.SA-PST-command 2.SM-visit-SUBJ 7-factory 7-all
'but the manager commanded them.'
(Ngonyani 1996: 76)

In this respect, Uyghur, European Portuguese, Greek and Swahili VSVE crucially differs from Hindi-Urdu, Japanese, Russian, and BCS VSVE. In the former languages verbal mismatches are possible in VSVE contexts; in in the latter languages they are impossible.

That VSVE languages differ regarding the identity conditions imposed on the stranded verb, can be stated in terms of whether (or not) the verbal identity requirement (VIR) holds for that language (see Goldberg 2005; Lipták 2012; Schoorlemmer and Temmerman 2012; Gribanova 2013b, 2020; McCloskey 2017; Fenger 2020, among others):

(44) **Verbal Identity Requirement**: In VSVE contexts, the antecedent and stranded main verb must be identical in root and derivational morphology.
(adapted from Goldberg 2005: 171)

Thus, given (44), while Hindi-Urdu (32), Japanese (33), Russian (34), and BCS (35) are VIR-obeying VSVE languages, Uyghur (36,37), European Portuguese (38,39), Greek (40,41), and Swahili (42,43) are not VIR-obeying VSVE languages.

As for what underlies whether the VIR holds or not in a given VSVE language, it has been proposed that the underlying mechanism is the type of head-dependency involved in V-stranding (McCloskey, 2017; Gribanova, 2020; Jenkins, 2025). It is well-known that ellipsis, quite generally, requires identity between the elided and antecedent phrases (Sag 1976; Chung et al. 1995; Merchant 2001, among others) and that movement of an element outside the ellipsis target can neutralize what otherwise would be an identity failure (Merchant 2001, among others). To illustrate, compare topicalization out of an VP-ellipsis target in (45a) with (45b).

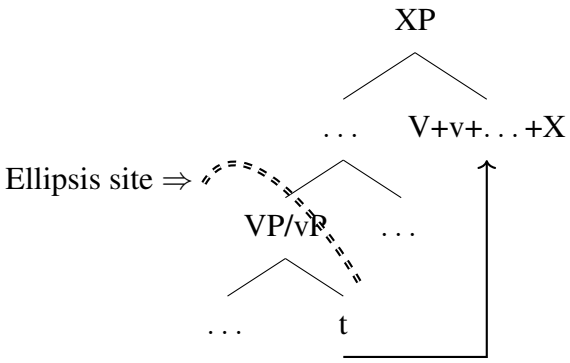
(45) a. Martha loves to eat ice cream, but pizza$_i$ she doesn't ~~love to eat t$_i$~~.
b. Martha loves to eat ice cream, but Peter doesn't ~~love to eat ice cream/*pizza~~.

That syntactic movement, as in (45a), can resolve what otherwise would be an identity failure under ellipsis, can be understood in terms of the lexical identity condition in (46) (see Chung 2006 and Merchant 2010 for further discussion of this condition):[6]

(46) Any non-trace morpheme *m* that occurs in an elided phrase must have an equivalent overt correlate *m′* in the elided phrase's antecedent.
(Merchant 2010: 55)

With respect to VSVE, it has been proposed that in the cases where the VIR does not hold, i.e. in cases like (36-43), the V has undergone syntactic head-movement from within the VP-ellipsis target to a higher functional project, as in (47) (Merchant 2018; Jenkins 2025).

(47)



Since in these cases the verb has undergone syntactic-head movement, the movement leaves a trace within the VP-ellipsis target, as in (47). Thus, since traces do not induce identity violations, per (46), the VP can be elided with a stranded verb that is distinct from the verb in the antecedent.

In VSVE languages where the VIR does hold, i.e. (32-35), the verb is displaced outside the ellipsis target as well. However, in such languages, it has been proposed that the type of head-dependency involved in verb displacement is PF movement— not syntactic head-movement (Schoorlemmer and Temmerman, 2012; McCloskey, 2017; Gribanova, 2020; Portelance, 2020;

[6]This is not to suggest that the syntactic identity condition in (46) is the only identity condition that is operative in ellipsis licensing and there are additional identity requirements as well (see, e.g., Merchant 2001, 2010). For present purposes, however, I will limit discussion to the syntactic identity condition in (46).

Jenkins, 2025). In such languages, V remains syntactically *in situ* inside the VP-ellipsis target and only undergoes movement in PF to a higher head outside the ellipsis target, as in (48). Crucially, since V only moves after syntactic identity, i.e. (46), is calculated for ellipsis, the VIR is enforced.[7]

(48) a. Syntax: $[_{XP}\ [_{\ldots}\ \ldots\ [_{VP/vP_{ellip.\ target}}\ \ldots\ V\ v]\ \ldots\ ]\ X\ ]$
b. PF: [[. . . [. . . V v] . . .] X] → [~~[. . . [. . .]]~~ V+v+. . . +X]

As for the trigger of PF V-movement in VIR-obeying VSVE languages, McCloskey 2017, Gribanova 2020, and Jenkins 2025 propose that in these contexts V undergoes movement in PF for to provide PF-support for an inflectional element that is outside the ellipsis site. For example, to provide PF-support for an affixal morpheme.

Thus, on these proposals the key difference between VSVE languages where the VIR is imposed and languages where it is not is the type of head-dependency that underlies V displacement. Namely, whether V displacement involves syntactic head-movement or PF movement. In the former verbal mismatches are allowed; in the latter they are not.

3 Extraction from ellipsis

I will now turn to the second type of variation, which concerns phrasal extraction out of VSVE targets. It is well-known that there is variation regarding whether Ā-extraction out of ellipsis targets is possible (Schuyler 2001; Aelbrecht 2010; Merchant 2010; Baltin 2012; van Craenenbroeck and Lipták 2010; Bošković 2014, among others). For example, English VP-ellipsis typically allows such extraction, as illustrated below with topicalization (49a) and *wh*-movement (49b,49c). In (49a) and (49b,49c), the VPs are the ellipsis targets and the VP-internal objects have undergone movement outside of the VP. Similarly, under sluicing, the TP is the ellipsis target and the *wh*-phrase undergoes movement outside the TP-ellipsis target, as in (50).

(49) a. Martha reads books, but, magazines$_i$ she doesn't $[_{VP}\Delta]$.
b. Which books$_i$ did Martha read t_i and which magazines$_j$ did Peter $[_{VP}\Delta]$?
c. Although, we don't know what$_i$ John read t_i, we do know what$_j$ Fred did $[_{VP}\Delta]$.
(adapted from Baltin 2012: 387)

(50) I think that Martha read something but I don't know what$_i$ $[_{TP}\ \Delta]$.

In contrast, there are cases where extraction out of the ellipsis target is barred. For example, Japanese clausal ellipsis quite generally disallows extraction out of CP-ellipsis target. As illustrated below, while scrambling out of a non-elided CP is possible (51a), scrambling out of an elided CP is impossible (51b).

[7]Note that an alternative analysis of the VIR effect has been proposed, where the VIR is due to independent prosodic factors. Namely, that in languages where the VIR holds, the stranded verb undergoes syntactic V-movement out of the ellipsis target but cannot bear a pitch accent (i.e. contrastive focus), thus, ruling out verbal mismatches under VSVE (see Merchant 2018 for discussion). However, Gribanova 2020 and Portelance 2020 show that this restriction cannot generalize to all VIR-obeying VSVE languages; that is, the existence of cases of VSVE where pitch accent can be placed on a stranded verb which is nevertheless VIR-obeying. Thus, the underlying cause of the VIR cannot simply be due to an independent prosodic requirement on verbs in VIR-languages.

(51) a. Sono hon-o_i Hanako-ga [$_{CP}$ Taroo-ga t_i katta to] omotteiru (koto)
that book-ACC Hanako-NOM Taroo-NOM bought C thinks fact
'That book, Hanako thinks that Taro bought'
(Saito 1992: 70)

b. *Sono hon-o_i Taroo-wa [$_{CP}$ Hanako-ga t_i katta to] itta ga,
that book-ACC Taroo-TOP Hanako-NOM bought C said though
zassi-o_j Ziroo-wa [$_{CP}$ Δ] itta.
magazine-ACC Ziroo-TOP said
'That book, Taro said that Hanako bought and, that magazine, Ziroo said (Hanako bought)'
(Saito 2007: 211)

Among VSVE languages there is variation regarding extraction as well. Interestingly, whether or not a VSVE language allows for extraction is not arbitrary. In the languages that permit phrasal extraction out of the VSVE target, the VIR can be violated. As shown in (52,53), in Uyghur internal arguments can be dislocated, i.e. scramble, out of the VSVE target.

(52) Aygül kiyimni asta-asta yudi, lekim chapanni$_i$, u Δ yumadi.
Ayghül dress.ACC slowly wash.PST.3SG, but jacket.ACC 3SG wash.NEG.PST3SG
'Ayghül washed the dress slowly, but as for the jacket, she didn't wash (slowly).

(53) Kitapni$_i$, Zemire t_i Ayghülga ettigende berdi, lëkin almani$_j$, u Δ
book.ACC Zemire Ayghül.DAT morning.LOC give.PST.3SG but apple.ACC 3SG
bermidi.
give.NEG.PST.3SG
'The book, Zemire gave to Ayghül in the morning, but the apple, she didn't give (to Ayghül in the morning).

Similarly, in European Portuguese, elements can be freely extracted from the VSVE target as well (see Santos 2009 for additional extensive discussion). As shown below, both *wh*-extraction (54a) and topicalization movement (54b) out of the VSVE target are possible.

(54) a. Eu entregui o teste aos alunos de segundo ano na segunda, e a quem$_i$ é
I gave the exam to.the students of second year on.the Monday and to whom is
que tu enregaste ~~o teste t_i na segunda~~?
that you gave the exam on.the Monday
'I gave the exam to second years students on Monday, and who did you give the exam to?'
(Santos 2009: 45)

b. À escola$_i$, ele leva sempre os filhos t_i e á natação$_j$ também leva Δ.
to-the school he takes always the kids and to-the swimming also takes
'To school, he always drives his kids, and to swimming, he does too.' (adapted from Martins 2016)

In Greek, extraction out of VSVE targets is also quite generally permitted (see Merchant 2018 for extensive discussion). As shown below, Greek VSVE allows for *wh*-extraction (55a) and topicalization (55b) out of VSVE targets.

(55) a. Me pjon$_i$ ithele i Maria na milisi t$_i$, ke me pjon$_j$ ithele i Ana Δ?
with whom wanted the Maria SUBJ speaks and with whom wanted the Ana
'With whom did Maria want to speak, and with whom did Anna want (to speak)?'
(Merchant 2018: 249)

b. Theno na miliso ja to proto thema. Ja to deftero$_i$, dhe thelo Δ.
want.1S SUBJ SPEAK.1S about the first topics about the second not want.1S
'I want to speak about the first topic. About the second, I don't want (to speak).'
(Merchant 2018: 259)

Finally, Swahili VSVE also allows for the same types of extraction as in the other VIR-violating VSVE languages. As shown in (56) topicalization out of the VSVE target is possible and (57) shows that *wh*-extraction is possible as well.

(56) Nyama$_i$, a-lii-pika vizuri t$_i$, lakini mboga$_j$ ha-ku-pika Δ.
meat, 1SM-PST-cook-FV well but vegetables NEG.1SM-NEG.PST-cook-FV
'The meat she cooked well, but the vegetables she didn't (cook well).

(57) a. a-li-sem-a kwamba a-ta-ib-a barua
1.SA-PST-say-FV that 1.SA-FUT-steal-FV 9.letter
'She said she will steal the letter.'

b. na ni-na-ju-a kwa nini a-ta-ib-a Δ.
and I-PRT-know-FV for what 1.SA-FUT-steal-FV
'and I know why she will'
(Ngonyani 1996: 83)

Now compare with Hindi-Urdu (58), Japanese (59), Russian (60), and BCS (61), which are all VIR-obeying languages (32-35). As shown below, in all of these languages Ā-extraction out of the VSVE target is not possible:

(58) *gaarii$_i$ Ram-ne dhyaan-se t$_i$ dho-ii lekin khirkiyaa$_j$ us-ne Δ nahiiN
car.F Ram-ERG carefully wash-PFV.F but window.F.PL he-ERG NEG
dho-ii.
wash-PFV.F.PL
'The car, Ram washed carefully, but the windows, he didn't (carefully) wash."

(59) *Bill-wa kuruma-o teinei aratta kedo, baiku$_i$-o kare-wa Δ arawanakatta.
Bill-TOP car-ACC carefully wash.PST but bike-ACC 3SG-TOP wash.NEG.PST
'Bill washed the car carefully, but, as for the motorbike, he didn't wash (carefully).

(60) *Tort$_i$, Marta ne prigotovila t$_i$ točno sleduja receptu, a pirog$_j$ Petja
cake.ACC Martha NEG prepare.SG.F exactly following recipe but pie.ACC Peter
prigotovil Δ.
prepare.SG.M
'The cake, Martha didn't prepare exactly following the recipe, but the pie, Peter prepared (exactly following the recipe).'

(61) *Šta$_i$ Marko piše t_i pažljivo, a šta$_j$ njegov asistent piše Δ?
what Marko writes carefully and what his assistant writes
'What does Marko write carefully and what does his assistant (write carefully)?'

Given the above facts, there is a striking correlation between VSVE languages where the VIR, i.e. (44), holds and whether XPs can be Ā-extracted out of the VSVE target. In all the VIR-obeying languages Ā-extraction out of the VSVE target is impossible and in all the VIR-violating languages Ā-extraction out of the VSVE target is possible. Thus, given this correlation, I propose the following generalization holds:

(62) **Extraction-VIR Generalization**: Ā-extraction out of the VSVE ellipsis target is only possible, if VIR-violations are possible.

4 Explaining the correlation

In the previous section, it was shown that there is a tight correlation between two otherwise, unrelated types of variation regarding ellipsis. Specifically, that there is a correlation between whether, for a given VSVE language, VIR violations are possible and whether Ā-extraction can proceed out of the VSVE target. In this section, I show both types of variation are regulated by the same underlying factor, namely, whether the VSVE target is a phase or a phase complement.

4.1 Blocking extraction

Previous works have proposed that ellipsis is phase constrained; that is, ellipsis can target either phase complements (Holmberg, 2001; Gengel, 2007; van Craenenbroeck, 2010) or full phases (Bošković, 2014; Lewis, 2024). That ellipsis can target phase complements can be shown with sluicing (63a) and possessor-stranding ellipsis (63b) in English. Assuming that CP and DP are phases and that sluicing involves elision of the TP complement and possessor-stranding ellipsis involves elision of the NP complement, then both cases are instances of elision of phase complements.

(63) a. Martha loves someone but I don't know [$_{CP}$ who [$_{TP}$ Δ]].
b. I read Pynchon's novel and you read [$_{DP}$ DeLillo's [$_{NP}$ Δ]].

Cases of full phase ellipsis are observed with Japanese clausal and argument ellipsis, as shown in (64) and (65), respectively. Assuming that CP and DP are phases in Japanese, that these phrases can be targeted for ellipsis indicates that full phase ellipsis is also an option (Bošković, 2014).

(64) Hanako-wa [$_{CP}$ zibun-no teian-ga saiyoosareru to] omotteiru ga, Taroo-wa Δ
Hanako-TOP self-GEN proposal-NOM accepted.be that think though Taro-TOP
omotte inai.
think not
'Hanako$_i$ thinks that her$_i$ proposal will be accepted, but Taro$_j$ does not think that her$_i$/his$_j$ proposal will be accepted.'
(Saito 2007: 203)

(65) Hanako-wa zibun-no koppu-o mitte kita; Taroo-mo Δ motte kita.
Hanako-TOP self-GEN glass-ACC brought Taroo-also brought
'Hanako brought her glass Taroo also brought her/his glass'
(Saito 2007: 205)

Bošković 2014 shows that with respect to extraction, phase complement and phasal ellipsis behave differently regarding Ā-extraction (see also Lewis 2024 for additional evidence). Namely, that while Ā-extraction is possible out of phase complement ellipsis targets, it is not out of a full phase ellipsis target. To illustrate, sluicing (e.g. (63a)) involves elision of a TP, i.e. the complement of the CP phase, plus *wh*-movement to Spec, CP. Conversely, as was shown and repeated below in (51b), Japanese clausal ellipsis involves elision of a full CP phase and Ā-extraction is impossible:

(51b) *Sono hon-o_i Taroo-wa [$_{CP}$ Hanako-ga t_i katta to] itta ga, zassi-o_j
that book-ACC Taroo-TOP Hanako-NOM bought C said though magazine-ACC
Ziroo-wa [$_{CP}$ Δ] itta.
Ziroo-TOP said
'That book, Taro said that Hanako bought and, that magazine, Ziroo said (Hanako bought)'
(Saito 2007: 211)

A similar split regarding extraction out of a phasal versus phase complement ellipsis target is observed in the nominal domain. As illustrated below in (66), in possessor-stranding ellipsis the NP phase complement is the ellipsis target and extraction of the PP is possible.

(66) ? [$_{PP}$About the Roman Empire$_i$] you told me to read [$_{DP}$ a historian's [$_{NP}$ research t_i]], but [$_{PP}$about the Great Depression]$_j$, you told me to read [$_{DP}$ an economist's [$_{NP}$ Δ.]]

However, in cases of argument ellipsis, i.e. elision of a full DP phase, the situation is different. Takahashi and Funakoshi 2013 shows that extraction of a *wh*-PP complement from inside a DP is possible in non-ellipsis contexts, e.g. in the response in (67c) . However, this type of extraction is not possible out of a DP-ellipsis target, as shown in (67d) (Sakamoto, 2020).

(67) a. A^1: Dare-kara$_i$-no Taroo-ga [t_i tegami]-o sute-ta no?
who-from-GEN Taroo-NOM letter-ACC discard-PST Q
'From who$_i$, Taro discarded [a letter t_i]?'

b. B: Bill-da yo.
Bill-PRES SFP
'Bill.'

c. A^2: Zyaa, dare-kara$_i$-no Ziroo-wa [$_{DP}$ t_i tegami]-o sute-ta no?
then who-from-GEN Ziro-TOP discard-PST Q
'Then, from whom$_i$ did Ziro discard a letter?'

d. A^3: *Zyaa, dare-kara$_i$-no Ziroo-wa [$_{DP}$ Δ] sute-ta no?
then who-from-GEN Ziro-TOP discard-PST Q
'Then, from whom$_i$ did Ziro discard?'
(Sakamoto 2020: 74)

Thus, as shown, there is a tight correlation between phase/phase complement ellipsis and whether Ā-extraction is possible.

Interestingly, predicate ellipsis contexts display the same kind of variability regarding extraction. That is, sometimes Ā-extraction is possible and in other cases it is not. As shown below, in typical VP-ellipsis contexts *wh*-extraction is possible:

(68) Which girls$_i$ did Martha invite t_i to the party and which boys$_j$ didn't she Δ?

The possibility of extraction in (68) is expected given that it involves a VP-ellipsis target (i.e. the complement of the vP phase), as is standardly assumed. However, there are also predicate ellipsis contexts that do not allow extraction. As shown in (69a), in contexts where predicate ellipsis occurs in a passive existential construction, extraction is not possible out of the ellipsis target.

(69) a. * I don't know what$_i$ there were so many getting worked up about yesterday, but I do know what$_j$ there were Δ last week.

b. I don't know what$_i$ there were so many people getting worked up about t_i.

(Harwood 2015: 535)

Harwood (2015) observes that in passive existentials, subjects cannot survive ellipsis quite generally, as in (70).

(70) John said there were several people arrested last night, and indeed there were (*several people) Δ.

(Harwood 2015: 535)

Assuming that in such constructions, the subject is in Spec, vP (Chomsky, 2001; Harwood, 2015), the impossibility of subjects surviving ellipsis indicates that vP is the ellipsis target. Thus, given the impossibility of extraction in (69a), this indicates that predicate ellipsis can also involve targeting the vP phase. More generally, what the above data indicate is that predicate ellipsis can target both the VP phase complement and the full vP phase.

To summarize, given the data in this section regarding the connection between extraction and the phasal status of the ellipsis target, the following holds (cf. Bošković 2014):

(71) If the XP-ellipsis target is a phase, then Ā-extraction out of XP is not possible.

Returning to VSVE, consider the data presented in the previous section. As was shown, there is a split among VSVE languages, where some languages allow extraction out of the ellipsis target and others do not. Further, this split regarding extraction correlated with whether the VIR held or not. Given the generalization above (71) regarding extraction out of ellipsis targets, I propose that the correlation regarding extraction out of a VSVE target and the VIR indicates that the following holds:

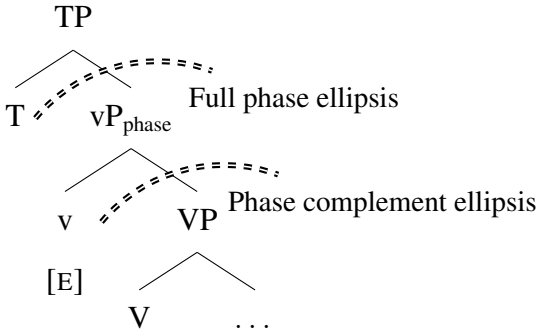
(72) If a VSVE language allows VIR violations, then the VSVE target is a phase complement.

4.2 Analysis

As was shown, in VSVE contexts, the possibility of Ā-extraction and whether the VIR holds are tightly correlated with one another. Moreover, it was shown that both possibilities correlate with the phasal status of the ellipsis target; that is, whether the ellipsis target is a phase or phase complement. If the VSVE target is a phase complement, then Ā-extraction and VIR violations are possible. But if the VSVE target is a phase, then Ā-extraction and VIR violations are not possible. In this section, I will propose an analysis that captures this correlation.

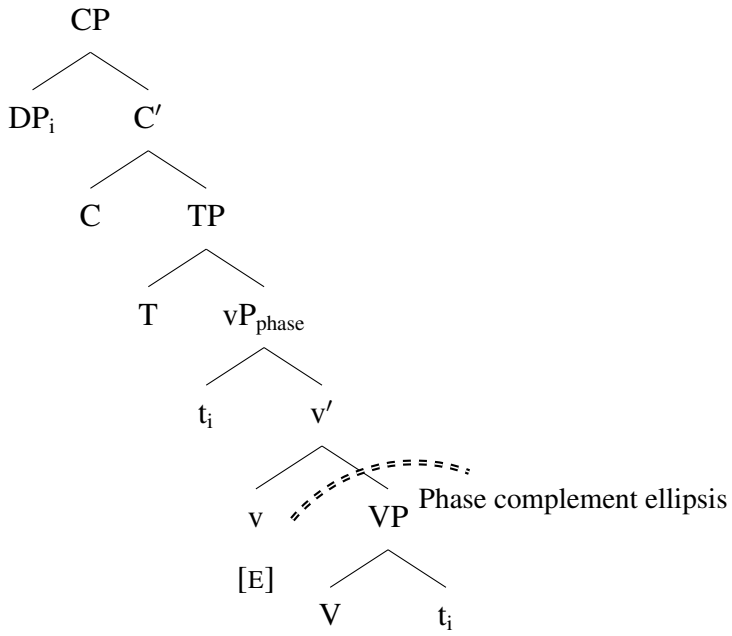
Given that both phase complement and full phase ellipsis are options, I adopt the proposal that predicate ellipsis involves an [E]-feature bearing v phase head, which marks either the full vP phase or its VP complement for ellipsis, i.e. zero-realized in PF, as in (73) (Bošković, 2014). Additionally, following standard assumptions (e.g. in Chomsky 2000, 2001), I assume that upon completion of the phase, i.e. after syntactic operations within the phase are complete, the material is sent to the interfaces, i.e. Spell-out, and is invisible to further movement operations. Importantly, following Bošković 2014, I adopt the proposal that the vP Spell-out domain can be either the VP phase complement or the full vP phase. With respect to ellipsis, when the VP is ellipsis-marked, the VP phase complement is the Spell-out domain; when the full vP phase is ellipsis-marked, the full vP phase is the Spell-out domain.

(73)



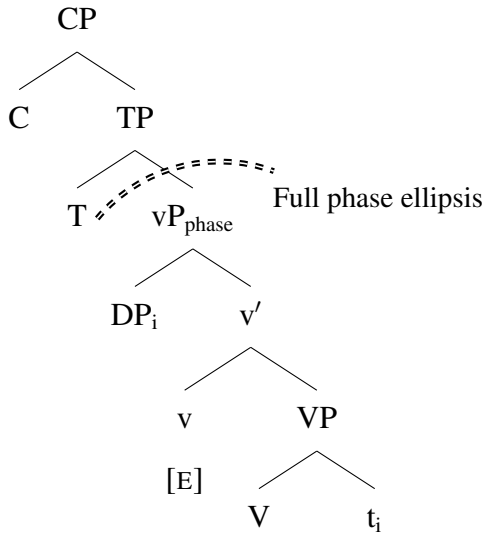
Now consider the blocking effect observed with extraction out of a phasal ellipsis target (but not with phase complement ellipsis targets). In the case of phase complement VP-ellipsis, upon the merger of the phase head v with VP, the DP undergoing Ā-movement to Spec, CP, moves to the phase edge, as an intermediate position (Chomsky, 2000, 2001). Then after movement operations are complete, the phase complement, i.e. VP, is sent to Spell-out, as in (74).

(74)



Compare with a derivation that involves full phase ellipsis, as in (75). As before, the phase head v merges with VP and the DP moves to Spec, vP. However, in this case, the entire vP phase is marked for ellipsis, which includes the DP in Spec, vP. Subsequently, upon the completion of the vP phase—i.e., after movement operations within the vP phase are complete, the full vP phase is sent to Spell-out, which includes the DP in Spec, vP. Thus, the DP is trapped and cannot undergo further movement.

(75)



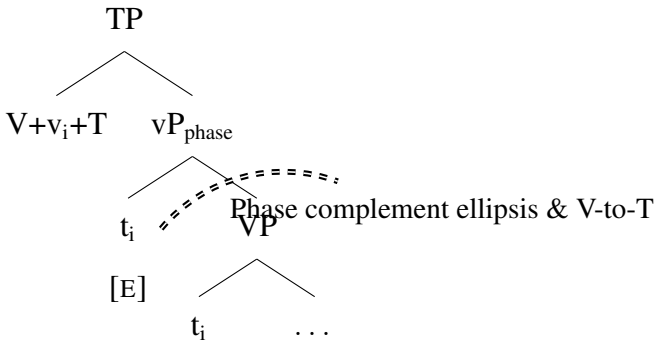
Thus, due to this difference regarding phasal versus phase complement ellipsis, there is a contrast in whether or not phrases are able to move out of the VSVE target.

Consider now the split among VSVE languages regarding the VIR, as was shown in section 3. In VIR-violating languages V undergoes syntactic V-to-T movement out of the ellipsis target, but in VIR-obeying languages V does not (it only undergoes movement in PF). I propose that this split

regarding whether syntactic V-movement occurs (or not) is due to the same type of blocking effect observed regarding phrasal movement.

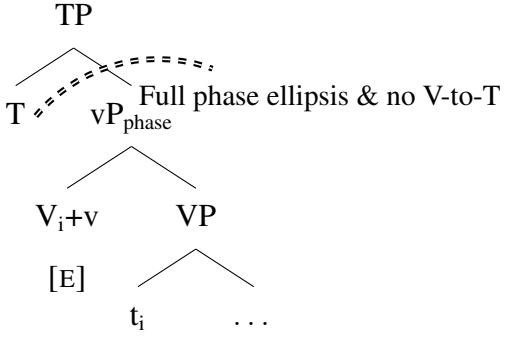
In VIR-violating languages, an [E]-bearing v merges with VP and V moves to v. After movement operations are complete, the VP phase complement is marked for ellipsis and sent to Spell-out. Since V is outside of the VP, it is not elided and can raise further to T, as in (74). Moreover, since V has undergone syntactic movement, thus, leaving a trace inside the ellipsis target, the VIR is not in effect and verbal mismatches are possible.

(76)



Compare with VIR-obeying VSVE, which involves full phase ellipsis. As before, v merges with VP and V moves to v. However, upon the completion of movement operations within the vP phase, the entire vP phase is marked for ellipsis and sent to Spell-out. Since, V is sent to Spell-out, it cannot undergo further movement upon the merger of T, as in (77a). However, in PF, since the inflectional T does not have a verbal host, I propose that V+v undergoes PF movement to support the inflection T as in (77b) (see Jenkins 2025 for discussion of this effect). Since PF movement, unlike syntactic movement, does not leave a trace, the V syntactically remains inside the vP-ellipsis target and, thus, the VIR holds.

(77) a.



b. PF: [. . . ~~[. . . V+v]~~ . . . T] → [. . . ~~[. . .]~~ . . . V+v+T]

On the present proposal, the correlation between Ā-movement out of the ellipsis VSVE target and the VIR are explained in terms of the same underlying mechanism. Namely, where the VSVE target is a full phase or a phase complement. In the case of full phase ellipsis, both Ā and X-movement are blocked due to the freezing effect that ellipsis marking of the vP phase induces. But in the case of phase complement ellipsis both Ā and X-movement are possible.[8]

[8]In light of the present proposal, it is useful to compare it with Landau's (2018; 2020; 2023) approach where all cases of VSVE are, in fact, cases of (multiple) argument ellipsis (or object *pro*). On Landau's approach, it is

5 Conclusion

To conclude, this paper has established a novel generalization that correlates two types of variation among VSVE languages: verbal identity under VSVE and the possibility of XP-extraction out of the VSVE target. I showed that if a VSVE language permits VIR-violations, then it also allows XP-movement of the ellipsis target. Further, I showed that what underlies this correlation was the phasal status of the ellipsis target. That is, whether VSVE involves full phase vP ellipsis or phase complement VP ellipsis. Additionally, I proposed an analysis that captures this correlation.

More generally, in this paper I have shown that whether a VSE language obeys the VIR is not an arbitrary point of variation. Rather, VIR effects, in fact, correlate with a more general type of variation observed among ellipsis phenomena, i.e. whether extraction is possible out of the ellipsis target. Specifically, I have shown that whether the V can undergo syntactic head-movement out of the vP in VSVE constructions is regulated by the same mechanism which underlies whether XP-movement out of an ellipsis target is possible: the phasal status of the ellipsis target. In contexts where the VSVE target involves full phase vP-ellipsis, neither syntactic V-movement nor XP-movement out of the vP are possible and the V must undergo PF movement, thus, inducing the VIR. But in contexts where VSVE involves elision of a phase complement, both syntactic V-movement and XP-extraction can proceed and the VIR may be violated. In this respect, VIR effects, in fact, may be a sub-case of a more general phenomena regarding extraction possibilities out of ellipsis, as the present analysis indicates.

Appendix: Going beyond VSVE with Verb echo answers

In a number of the languages discussed, in addition to allowing VSVE, those languages also allow another, similar, ellipsis construction: verb echo answers (VEA) (Holmberg, 2015). For example, both Japanese (Sato and Hayashi, 2018; Sato and Maeda, 2021) and Hindi-Urdu (Dash, 2020) allow VEA constructions, as in (78b) and (79b), respectively.

(78) a. Moo tuki-masi-ta-ka?
already arrive-POL-PAST-Q
'Did you arrive already?'

b. Tuki-masi-ta-yo.
arrive-POL-PAST-PRT
Intended: 'I arrived already.'
(Sato and Maeda 2021: 361)

highly unexpected that Ā-extraction is blocked in VIR-obeying VSVE contexts, given that on Landau's approach there is no phasal vP-ellipsis target to block extraction (rather argument(s) are elided and the vP/VP is not elided), thus, Ā-extraction should, in principle, be possible in VIR-obeying VSVE contexts by simply not eliding the extracted argument. Moreover, that whether Ā-extraction is (im)possible in VSVE contexts correlates with whether the VIR is obeyed or not (i.e. (62)) is also unexpected on this account. On Landau's account since all cases of VSVE are uniformly the result of argument ellipsis (or object *pro*), VIR effects must be explained in terms of a mechanism independent of argument ellipsis (or object *pro*). Since argument deletion and the VIR would be due to distinct processes on Landau's account, it is not straightforwardly predicted that the VIR and the impossibility Ā-extraction should be correlated (in contrast to the present account, where whether the VIR holds and whether Ā-extraction is possible straightforwardly follows from a single underlying process— i.e. whether the VSVE target is a phase or phase complement).

(79) a. (kyaa) raaghav-ne phirse kitaab paDhii?
Q Raghav-ERG again book read.PERF.3.F.SG
'Did Raghav read the book again?'

b. haaN paDhii.
yes read.PERF.3.F.SG
'Yes. (Raghav performed the action of reading the book again.)'
(Dash 2020: 7)

As illustrated in (78b,79b) above, VEA can be characterized as occurring in polar question response contexts. In such contexts, the verb in the question is repeated in the response and is interpreted as giving an answer to the question. Importantly, VEA has been argued to also involve phrasal ellipsis. As (78b,79b) illustrate, the VEAs involve a manner-adjunct interpretation, indicating that VEAs involve elision of a phrase that contains these adjuncts (and is not due to *pro* arguments or argument ellipsis; see Sato and Hayashi 2018; Dash 2020; Sato and Maeda 2021).

While VEA bears a similarity to VSVE, it differs regarding the availability of overt subjects in such contexts. As shown below in (80b,80c) and (81b,81c), overt subjects are not possible in VEA contexts.

(80) a. Q: Dareka-ga kinoo kokode tabako-o sui-mashi-ta-ka?
someone-NOM yesterday here cigarette-ACC smoke-POL-PAST-Q
'Did anyone smoke a cigarette here yesterday?'

b. A: Sui-mashi-ta-yo. Tabun Fujita-san-desu-ne.
smoke-POL-PAST-PRT probably Fujita-TIT-COP.POL-PRT
'Yes. Someone smoked a cigarette yesterday. Probably it was Fujita.'
(Sato and Hayashi 2018: 76)

c. A': *Dareka-ga sui-mashi-ta-yo. Tabun Fujita-san-desu-ne.
someone-NOM smoke-POL-PAST-PRT probably Fujita-TIT-COP.POL-PRT
Intended 'Yes. Someone smoked a cigarette yesterday. Probably it was Fujita.'

(81) a. Q: (kyaa) kisi-ne aasani-se kaam kaiyaa?
Q someone-ERG easily-WITH work do
'Did someone do the work easily?'

b. A: Haan, kaiyaa.
yes do
'Yes. Someone did the work easily.'

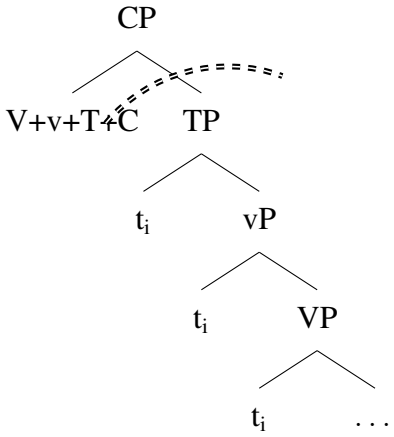
c. A': *Haan, kisi-ne kaiyaa.
yes someone-ERG do
'Yes, someone did the work easily.'

That VEA disallows overt subjects has been interpreted as indicating that the VEA construction involves TP-ellipsis, which includes the subject in Spec, TP (Holmberg, 2015; Gribanova, 2017; Sato and Hayashi, 2018; Dash, 2020; Sato and Maeda, 2021).[9] On this proposal, V undergoes

[9]Note that an analysis of VEA, that involve a null subject *pro* can be ruled out due to the general ban on indefinite subject *pro* (Holmberg, 2015). Since the VEAs in (80b) and (81b) involve null indefinite subjects, this indicates that the subject is not *pro* in these constructions.

movement above TP to C and the TP is elided, as in (82) below:

(82)



Interestingly, VEA differ from VSVE in another respect regarding the VIR. As was shown in section 2, Japanese and Hindi-Urdu were both VIR-obeying VSVE languages, where the stranded verb must be lexically identical to the verb in the antecedent. However, in VEA contexts, the verb in the response can be distinct from the verb in the antecedent question, as in (83) and (84).

(83) a. Q: Gozira-ga issyunnoutini biru-o **humitsubusi**-ta-no?
Godzilla-NOM in.an.instant building-ACC trample-PST-Q
'Did Godzilla trample down the building in an instant?'

b. A: (Iya,) **moyasi**-ta-yo.
no burn-PST-PRT
'Godzilla burned down the building in an instant.'
(Tanabe and Kobayashi 2023: 469)

(84) a. Q: Kyaa Mary-ne asaanii-se rotii **banayii**?
Q Mary-ERG easily-with bread cook.PERF.F
'Did Mary make bread easily?'

b. A: Nahi, **belii**.
no rolled.PERF.F
'No, she rolled it easily.'

That VEA permit verbal mismatches, even in languages that obey the VIR for VSVE, I argue is due to VEA involving TP-ellipsis; that is, ellipsis of the complement of the CP phase. Given the account of ellipsis developed in section 4, I propose the in VEA, upon the merger of the phase head C, V+v+T raises to C and the TP phase complement is marked for ellipsis. Upon completetion of syntactic operations in the CP phase, the TP is sent to Spell-out. Since V has undergone syntactic movement outside of the TP, verbal mismatches are possible.

Given that VEA involves phase complement ellipsis, unlike VSVE, which involves full phase ellipsis in languages like Hindi and Japanese, we should expect that extraction is possible out of the TP-ellipsis target per the generalization in (71). As shown below, this is indeed the case, in both Japanese (85b) and Hindi (86b) VEAs, objects can be extracted out of the TP-ellipsis target:

(85) a. Q: Gozira-ga issyunnoutini biru-o humitsubusi-ta-no?
Godzilla-NOM in.an.instant building-ACC trample-PST-Q
'Did Godzilla trample down the building in an instant?'

b. A: ?Iya Kuruma$_i$-o, Δ humitsubusi-ta-yo.
no car-ACC trample-PST-PRT
'No, the car he did in an instant.'

(86) a. Q: Kyaa Mary-ne asaanii-se car dhoyii?
Q Mary-ERG easily-with car wash.PERF.F
'Did Mary wash the car easily?'

b. A: Nahi, par motorcycle$_i$ Δ dhoyii.
no but motorcycle wash.PERF.F
'No, but the motorcycle she did wash easily.'

That VEA, unlike VSVE in Japanese and Hindi-Urdu, allow for both verbal mismatches and extraction, further confirm the correlation between the phasal status of the ellipsis target and the possibility of verbal mismatches.

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